PIPING THICKNESS MONITORING SURVEY

Crude Distillation Unit - Circuit CDU-P-11, Survey Round 2024

Survey No.: TMS/2024/CDU/03
Survey Date: 26 Feb 2024
Circuit: CDU-P-11
Line: 8-P-1104
Technician: M. Fernandes, UT Level II
Instrument: DMS Go, Sl. 88213, cal. 04 Jan 2024

1. Circuit Description

Circuit CDU-P-11 comprises the 8 inch overhead transfer line 8-P-1104 from the atmospheric column overhead to the first stage condenser, including four elbows, one reducer and the associated branch connections. The circuit operates in wet hydrogen sulphide service and is subject to a documented corrosion mechanism of ammonium chloride under-deposit attack.

The retirement thickness for line 8-P-1104 is 5.2 mm, derived from the pressure design thickness of 4.4 mm plus a structural minimum margin. Readings approaching this value require immediate referral to the Inspection Engineer.

2. Measurement Results

CML	Component	Nominal	Previous	Current	Rate (mm/yr)	Remaining Life
CML-01	Straight run A	8.18	7.95	7.91	0.04	> 20 yr
CML-02	Elbow E1 extrados	8.18	7.62	7.50	0.12	19.2 yr
CML-03	Elbow E1 intrados	8.18	7.41	7.26	0.15	13.7 yr
CML-04	Straight run B	8.18	7.88	7.83	0.05	> 20 yr
CML-05	Tee T1 branch	8.18	7.55	7.44	0.11	20.4 yr
CML-06	Tee T1 run	8.18	7.71	7.64	0.07	> 20 yr
CML-07	Reducer R1 large end	8.18	7.33	7.19	0.14	14.2 yr
CML-08	Reducer R1 small end	6.35	5.98	5.90	0.08	8.8 yr
CML-09	Straight run C	8.18	7.90	7.86	0.04	> 20 yr
CML-10	Elbow E2 extrados	8.18	7.68	7.59	0.09	> 20 yr
CML-11	Elbow E2 intrados	8.18	7.29	7.14	0.15	12.9 yr
CML-12	Straight run D	8.18	7.92	7.80	0.12	21.7 yr
CML-13	Dead leg to drain	8.18	6.98	6.79	0.19	8.3 yr
CML-14	Straight run E	8.18	7.87	7.81	0.06	> 20 yr
CML-15	Elbow E3 extrados	8.18	7.55	7.47	0.08	> 20 yr
CML-16	Elbow E3 intrados	8.18	7.18	7.02	0.16	11.4 yr
CML-17	Straight run F	8.18	7.84	7.79	0.05	> 20 yr
CML-18	Condenser inlet nozzle	8.18	7.44	7.35	0.09	> 20 yr
CML-19	Low point drain boss	6.35	5.86	5.62	0.24	1.8 yr
CML-20	Vent connection	6.35	6.02	5.97	0.05	15.4 yr

3. Corrosion Rate Assessment

CML	Short Term Rate	Long Term Rate	Governing Rate	Assessment
CML-12	0.12 mm/yr	0.09 mm/yr	0.12 mm/yr	Acceptable, continue monitoring
CML-13	0.19 mm/yr	0.16 mm/yr	0.19 mm/yr	Elevated, dead leg to be removed
CML-16	0.16 mm/yr	0.13 mm/yr	0.16 mm/yr	Acceptable, annual monitoring
CML-19	0.24 mm/yr	0.21 mm/yr	0.24 mm/yr	Lowest remaining life in circuit

CML-19 at the low point drain boss shows the lowest remaining life in the circuit at 1.8 years and governs the next inspection interval for the whole of CDU-P-11. A repair or replacement plan is to be raised before the next turnaround.

Surveyed By	Reviewed By	Approved By